

A Proposal for an Ensemble of Clustering Models for High-Dimensional Data

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Abstract—High-dimensional datasets present significant challenges for clustering, including redundant and irrelevant features, sparsity, noise, and the curse of dimensionality. Traditional clustering algorithms often struggle under these conditions, especially in resource-constrained environments. This work proposes an adaptive ensemble clustering framework specifically designed for high-dimensional data, integrating unsupervised feature selection (Infinite Feature Selection, Inf-FS), random projections for dimensionality reduction, subsampling for data diversity, and a diverse suite of clustering algorithms and consensus functions. The methodology is systematically benchmarked across a range of heterogeneous datasets—including spectroscopy, genomics, and image analysis—demonstrating improvements in clustering robustness, stability, and accuracy, especially for data with high feature redundancy. Experimental results show that the impact of feature selection and dimensionality reduction varies across datasets, highlighting the value of a flexible and interpretable ensemble design. The framework is computationally efficient and reproducible, enabling scalable deployment across a range of data sizes and computing environments. The study also underscores the need for broader dataset evaluation and the integration of explainable AI techniques to enhance the interpretability and transparency of clustering results, while providing a flexible foundation for future research on scalable and generalizable clustering in high-dimensional settings.

Index Terms—Ensemble clustering, high-dimensional data, unsupervised feature selection, random projections.

I. INTRODUCTION

CLUSTERING plays an essential role in data analysis as an unsupervised machine learning technique that seeks to group data into meaningful clusters, where elements within the same group share greater similarity compared to those in other groups. This technique has broad applications in diverse fields, including medicine, biology, civil engineering, market research, and social sciences, among others. Despite its utility, it remains a challenging problem due to the absence of ground truth labels, the sensitivity of algorithms to parameter choices, the wide variety of data distributions, and the difficulty of determining the most appropriate number of groups. Moreover, finding optimal clustering solutions is considered an NP-hard problem, which further increases complexity [1].

A significant theoretical challenge in clustering is presented in Kleinberg’s Impossibility Theorem [2]. This theorem asserts that no clustering algorithm can simultaneously satisfy three desirable properties: scale-invariance (the clustering does not change if all distances are multiplied by a positive constant), richness (every possible partition of the data can be obtained by the algorithm for some distance function), and consistency

(if distances within clusters are decreased and distances between clusters are increased, the clustering does not change). This result underscores the inherent difficulty of achieving global optimal clustering solutions, particularly when dealing with real-world data that do not adhere to ideal assumptions.

The challenges of clustering are intensified in high-dimensional spaces due to the “curse of dimensionality” [3]. In this scenario, data points tend to become sparse, and the concept of distance, a fundamental aspect of many clustering algorithms, becomes less meaningful [4]. As the number of dimensions increases, the relative distance between the nearest and farthest points diminishes, leading to poor discrimination of distances. This situation not only complicates the identification of meaningful clusters but also increases computational complexity, making traditional clustering algorithms less effective.

Additionally, high-dimensional data often include irrelevant or noisy features that can confound clustering algorithms. For example, algorithms dependent on Euclidean distance may struggle to differentiate between relevant and irrelevant dimensions, further degrading clustering performance [5]. These issues highlight the importance of identifying relevant features and employing advanced preprocessing techniques, such as feature selection and dimensionality reduction, to improve clustering outcomes.

To address these challenges, ensemble clustering has emerged as a promising approach. This method refers to the process of combining multiple clustering results into a single consensus partition, leveraging the strengths of individual algorithms while mitigating their weaknesses. The key idea is to achieve greater robustness, stability, and accuracy by aggregating diverse clustering results [6]. This approach is especially beneficial in high-dimensional settings, where different algorithms may capture complementary aspects of the data [7]. Furthermore, ensemble methods can incorporate feature selection and dimensionality reduction techniques to generate diverse base clusterings and explore various data subspaces, ultimately enhancing the robustness and effectiveness of the final clustering solution [5].

This study introduces an ensemble clustering framework specifically designed for high-dimensional data. A novel contribution of this work lies in the integration of unsupervised feature selection, dimensionality reduction techniques, subsampling for data diversity, and the generation of diverse base clusterings using multiple algorithms and consensus functions.

By leveraging these components, the framework addresses the challenges of sparsity, noise, and computational complexity inherent in high-dimensional data, ultimately delivering a robust and scalable clustering solution.

To evaluate its effectiveness, a wide array of ensemble configurations is systematically tested across high-dimensional benchmark datasets. The findings show that combining these components improves clustering performance on challenging datasets. However, the impact of these techniques is data-dependent, highlighting the importance of adaptive ensemble design and rigorous benchmarking for optimal performance. This systematic approach, supported by a reproducible experimental setup, represents a significant advancement over previous studies that typically evaluated only isolated components or limited combinations of ensemble methods.

This study is grounded in an extensive and systematic review of the state of the art in clustering methods for high-dimensional data, covering feature selection techniques, dimensionality reduction, base clustering generation strategies, consensus functions, and validation metrics, among others. This comprehensive review not only highlights current limitations and challenges, but also provides the conceptual foundation and motivation for the adaptive ensemble framework proposed in this study.

The rest of the work is structured as follows. Section II provides a comprehensive review of the state of the art, discussing topics such as feature selection, dimensionality reduction techniques, generation of base clusterings, ensemble selection, consensus functions, validation metrics and evaluation. Section III introduces the proposed solution, an ensemble of clustering models designed specifically for high-dimensional data. Section IV presents the experimental results and evaluation, including comparisons with existing methods and analysis of the solution’s performance. Finally, Section V concludes the work by summarizing the findings and outlining potential directions for future research.

II. REVIEW OF THE STATE OF THE ART

The application of ensemble methods in clustering high-dimensional datasets has gained significant attention in recent years due to their potential to improve clustering accuracy, robustness, and stability. These methods leverage multiple clustering solutions to derive a consensus clustering that addresses the inherent challenges of high-dimensional data, such as sparsity, noise, and irrelevant and redundant features.

The concept of cluster ensembles as a knowledge reuse framework, which enables the combination of multiple clustering solutions into a single, robust consensus, was introduced in [6]. A key advantage of this approach is its ability to operate without requiring access to the original data features or the algorithms that produced the initial clusterings; instead, it relies solely on the cluster labels provided by each clustering solution. Three consensus functions—Cluster-based Similarity Partitioning Algorithm (CSPA), HyperGraph Partitioning Algorithm (HGPA), and Meta-CLustering Algorithm

(MCLA)—were proposed and demonstrated improved clustering quality and robustness across various datasets, including high-dimensional ones. The use of hypergraph representations facilitates effective integration of multiple clusterings, and the ensemble approach can help mitigate some challenges associated with high-dimensional data.

The method introduced in [8] for consensus clustering applies subsampling (sampling without replacement, 80% of the items from the original data) to generate multiple perturbed datasets from the original data. By applying a clustering algorithm repeatedly to these perturbed datasets, the method constructs a consensus matrix that quantifies the proportion of times each pair of items co-cluster across runs. This matrix serves as a stability measure of the clusters, providing insights into cluster number, membership, and boundaries. The approach is model-independent and mitigates issues related to sensitivity to initial conditions and sampling variability, making it particularly useful for complex, noisy, and small-sample, high-dimensional datasets.

Probabilistic and statistical approaches for clustering ensembles, including a mixture model of multinomial distributions for consensus clustering, were explored in [9], [10]. Ensemble methods can successfully combine weak clusterings—such as those derived from random subspaces, projections, or random hyperplane splits—to produce a superior overall clustering solution. While the mixture model approach requires specifying parameters like the target number of clusters and its performance can be influenced by the diversity and resolution of base clusterings, empirical results show that even simple or weak base clusterings can yield strong consensus results when properly combined.

A bipartite graph partitioning approach for combining cluster ensembles was introduced in [11], demonstrating improved and robust clustering results. To generate diverse base clusterings for high-dimensional data, random projections were used as one ensemble generation method, providing different views of the data and increasing clustering diversity. Although this approach enhances ensemble diversity, both diversity and quality were noted to impact final performance, and highly diverse ensembles (such as those from random projection) may present challenges for some methods that rely on correspondence between clusters.

The Random Projection Ensemble Clustering (RPEClu) [12] algorithm is a model-based ensemble algorithm specifically designed for high-dimensional data. RPEClu employs random projections to reduce the dimensionality of the data, thereby simplifying the estimation of Gaussian mixture models (GMMs), which can be computationally challenging in high dimensions. The algorithm applies GMM clustering to several random low-dimensional projections of the data, and a statistical selection process—such as evaluating the Bayesian Information Criterion (BIC)—is used to retain only the most informative projections. These selected clusterings are then aggregated to form a consensus solution, increasing stability and resilience against noise and overfitting. This method is particularly effective for datasets where conventional GMMs

struggle due to the “curse of dimensionality,” and it demonstrates improved performance, especially when the original variables are highly correlated or when clusters have complex shapes. The computational complexity of RPEClu is $O(Bm^3)$, where m is the number of original features of the dataset and B is the number of random projections applied by the algorithm.

Weighted cluster ensemble methods that assign feature-level weights within clusters produced by subspace clustering algorithms were proposed in [7]. This approach improves ensemble performance by leveraging the diversity of clusterings generated with varying parameters and by embedding feature relevance information into the consensus process. The study highlights the challenge of parameter selection in subspace clustering and demonstrates that ensemble methods can provide robust and accurate consensus clusterings without requiring prior knowledge or validation of input parameters.

Recent research has increasingly focused on the scalability of ensemble clustering methods. As discussed in the survey [13], advanced techniques such as graph-based and pairwise similarity-based approaches have been developed to combine clustering results efficiently while preserving solution quality. The survey highlights the strengths of ensemble clustering in integrating heterogeneous data types and providing robust solutions in the face of data perturbations, attributes that have proven particularly valuable in fields like bioinformatics and cybersecurity. However, the authors acknowledge that scalability remains a significant challenge, especially for similarity-based consensus functions, which often have quadratic complexity with respect to the number of data points.

Practical recent applications of ensemble clustering include the study [14] introducing an ensemble clustering method for assessing air quality monitoring networks in Mexico, showcasing the versatility of these methods in environmental science. This approach combined clustering ensembles to evaluate pollution patterns in major metropolitan areas, highlighting their adaptability to real-world datasets. Specifically, the authors applied principal component analysis, hierarchical clustering, and k-means within an ensemble framework to identify similar and redundant monitoring stations in Mexico’s three largest cities. This methodology not only improved the robustness of network assessment but also provided actionable insights for optimizing air quality monitoring, underlining the value of ensemble clustering in complex environmental applications.

Ensemble clustering methods have become a promising approach to address the challenges of high dimensionality frequently associated to bioinformatics data. An example is the single-cell RNA sequencing (scRNA-seq) data analysis [15]. In scRNA-seq, identifying cell types and understanding cellular heterogeneity are critical, but the data are often complex, sparse, and noisy, which can undermine the stability and accuracy of individual clustering algorithms. By generating multiple clustering partitions—through varying gene features, cell samples, or clustering algorithms—and integrating them using strategies such as voting or hypergraph-based aggregation, ensemble clustering leverages the complementary

strengths of different methods and compensates for their individual weaknesses. This integration yields more robust, accurate, and stable clustering outcomes, thereby enhancing the identification of cell types and the interpretation of cellular heterogeneity in scRNA-seq datasets.

A. Clustering Ensemble Framework

Clustering ensemble methods seek to combine multiple clustering results into a single, superior partition. Several properties have been proposed as desirable criteria for the effectiveness and reliability of clustering ensemble algorithms [10], [16], [17]:

- **Robustness:** The ensemble should outperform or, at minimum, match the average performance of individual clustering algorithms, especially in the presence of noise and outliers.
- **Consistency:** The consensus partition should be similar to the individual clusterings being combined, reflecting the structural agreement among base clusterings.
- **Novelty:** The ensemble method should be able to discover clustering solutions that are unattainable by any single base clustering algorithm alone, potentially revealing new data structures.
- **Stability:** The resulting clustering should exhibit low sensitivity to variations in the data, parameter settings, and algorithmic randomness, providing reliable results across runs.

The general pipeline for clustering ensemble methods consists of six key steps [13], [18]–[20], as illustrated in Figure 1:

- 1) **Preprocessing and Feature selection:**

The initial step involves identifying and selecting the most relevant features from the high-dimensional dataset, possibly utilizing unsupervised feature selection techniques to reduce noise and irrelevant information. This may also include standard data preprocessing steps such as normalization and handling missing values.

- 2) **Dimensionality Reduction:**

To further mitigate the curse of dimensionality, dimensionality reduction techniques such as Principal Component Analysis (PCA) or random projections methods are applied. These techniques transform the data into lower-dimensional subspaces while preserving as much structural information as possible.

- 3) **Generation of Base Clusterings:**

Multiple base clustering solutions are generated by applying different clustering algorithms, or the same algorithm with varying parameters (e.g., number of clusters, initialization, distance metrics), on the original or reduced data. Diversity in base clusterings can also be achieved by subsampling data points or features, or by projecting the data onto random subspaces.

- 4) **Ensemble Selection:**

This is often a beneficial step in the clustering ensemble process, particularly when the pool of base clusterings is large or heterogeneous in quality and diversity. Rather

than combining all generated base clusterings, ensemble selection aims to identify and retain a subset of base clusterings that are both high-quality and diverse, thus optimizing the performance of the final consensus solution.

5) **Consensus Function:**

The ensemble framework then combines the diverse set of base clusterings into a single consensus partition using a consensus function. Popular approaches include re-labeling and voting-based, co-association/pairwise similarity matrix, graph and hypergraph-based, and median partition / optimization-based, each aggregating the input clusterings in different ways to derive a robust final solution.

6) **Validation Metrics and Evaluation:**

The final clustering solution is evaluated using internal and/or external validation metrics, such as Silhouette Coefficient, Dunn Index, Adjusted Rand Index (ARI), or Normalized Mutual Information (NMI). These metrics assess the quality, stability, and reliability of the consensus clustering, guiding the selection of optimal configurations and ensemble parameters.

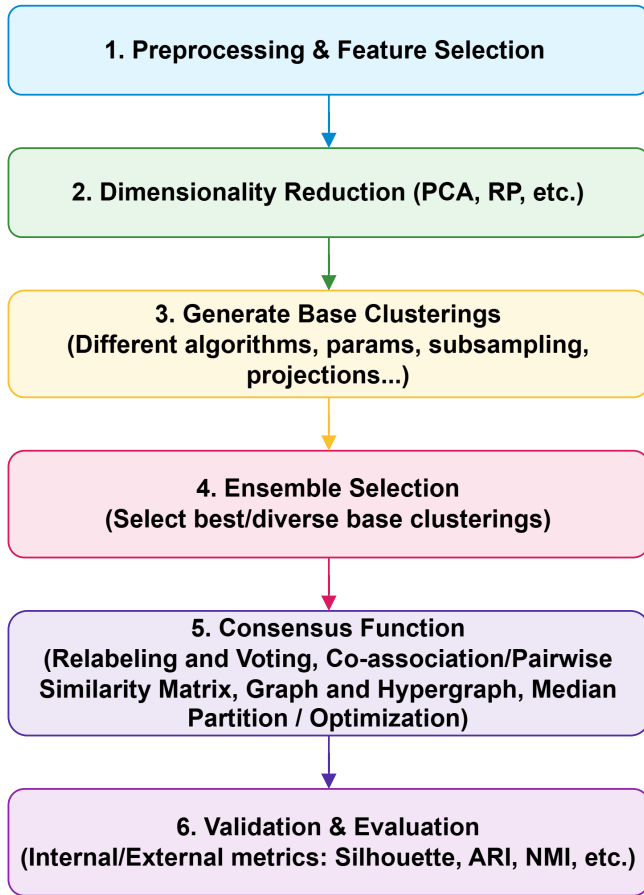


Fig. 1. General pipeline for clustering ensemble methods

The following subsections provide a detailed overview of the key aspects involved in the clustering ensemble process.

B. Unsupervised Feature Selection

Unsupervised feature selection aims to identify the most relevant features of high-dimensional data without relying on labeled examples. It is a critical step in unsupervised learning and clustering tasks, as it improves model interpretability and reduces computational costs. Unlike supervised feature selection, where class labels guide the search, unsupervised methods must define “interestingness” or “relevance” directly from the data. Here, “interestingness” refers to how well a subset of features uncovers meaningful cluster structure according to a chosen criterion, while “relevance” denotes the contribution of individual features toward revealing these structures in the absence of class labels [21]. Unsupervised feature selection methods can generally be classified into three main categories: filter, wrapper, and embedded approaches, although some methods incorporate hybrid combinations of these primary types.

Filter methods evaluate features based on intrinsic properties of the data, without relying on the outcome of clustering or classification algorithms. They are fast and scalable, making them suitable for large datasets. These methods often use statistical measures or geometric criteria to assess the importance of features. For example, the Laplacian Score [22] evaluates features by their ability to preserve the local manifold structure of the data. This method constructs a nearest neighbor graph to model the local geometric relationships among data points and computes, for each feature, a score that quantifies how well the feature maintains this locality structure. Features with lower Laplacian Scores are considered more relevant for clustering tasks, as they better preserve locality information and are more likely to capture the underlying intrinsic structure of the data.

Infinite Feature Selection (Inf-FS) [23] is a graph-based feature selection algorithm that ranks and selects features in unsupervised settings. Inf-FS represents feature subsets as paths in a fully-connected weighted graph, where each node corresponds to a feature and each edge models a pairwise relationship reflecting relevance and redundancy between features. The strength of these relationships is encoded in the adjacency matrix, which can be defined using various criteria—such as variance and correlation for Inf-FS.

To evaluate the “importance” of each feature, Inf-FS considers all possible paths (i.e., feature subsets) of arbitrary lengths within the graph. By leveraging properties of matrix power series and concepts from Markov chains, Inf-FS efficiently computes a feature relevance score by summing the contributions of all paths containing a particular feature, extending the analysis to infinite path lengths. This infinite-path approach allows Inf-FS to efficiently capture high-order dependencies among features (i.e., beyond simple pairwise interactions), while keeping the computational complexity manageable. The final feature ranking is derived from these scores: features that are both individually relevant and collectively non-redundant (i.e., those consistently appearing in high-value paths) are ranked highest.

A significant challenge in feature selection is determin-

ing how many features to keep. Inf-FS addresses this with an automatic subset selection mechanism. After scoring all features, Inf-FS analyzes the distribution of feature scores and applies a clustering algorithm (such as 1D Mean-shift) to partition features into groups, typically separating highly relevant features from less informative or redundant ones. The cluster containing the most relevant features determines the subset to retain. The computational complexity of Inf-FS is $O(m^3(1+n))$, where m is the number of features and n is the number of samples.

Wrapper methods integrate feature selection directly with clustering algorithms, evaluating feature subsets based on how well they uncover meaningful cluster structure. Although more computationally intensive than filter or embedded methods, wrappers like Feature Subset Selection using Expectation-Maximization clustering (FSSEM) [21] often yield superior results by systematically searching feature subsets and evaluating each via clustering performance criteria such as scatter separability or maximum likelihood. FSSEM uniquely addresses key challenges in unsupervised feature selection: it determines the optimal number of clusters concurrently with feature selection and corrects for biases related to feature subset dimensionality, making it especially robust for high-dimensional data clustering tasks.

Embedded methods integrate feature selection into the learning model itself, allowing simultaneous optimization of feature selection and the learning of clustering or dimensionality reduction structures. This approach leverages the strengths of machine learning models to identify relevant features during training, making them both efficient and effective for unsupervised tasks such as clustering or dimensionality reduction. An example is the Neural Networks and Self-Expression (NNSE) method [24], which combines neural networks and self-expression models in a unified framework. In the context of feature selection, self-expression refers to modeling each feature as a linear combination of all the features in the original space, where the learned weights indicate the importance and redundancy of each feature. By minimizing the reconstruction error, the self-expression model highlights the most representative features while filtering out redundant ones. NNSE replaces traditional linear spectral analysis with neural networks to learn nonlinear relationships between data and pseudo labels, uses self-expression to explore feature relationships and select representative features, and employs adaptive graph regularization to preserve the local manifold structure of the original data.

Mean correlation among features quantifies the average pairwise linear relationship between variables in a dataset, making it a simple and effective measure for understanding feature redundancy. It serves as an indicator of redundancy: a high mean correlation suggests many features provide overlapping information, while a low mean correlation implies relatively diverse and less redundant features. Assessing mean correlation before applying feature selection methods can guide expectations about the potential benefits of feature selection. When mean correlation is high, feature selection

primarily helps by removing redundant features, reducing dimensionality, and mitigating noise, which can substantially enhance clustering performance. Conversely, if mean correlation is already low, indicating diverse features, the redundancy reduction benefit of feature selection may be limited.

C. Dimensionality Reduction Techniques

Dimensionality reduction is a critical preprocessing strategy for managing high-dimensional datasets, particularly in unsupervised learning scenarios. It involves transforming features from the original dataset to reduce its dimensionality while retaining as much relevant information as possible [25]. This process helps mitigate issues like the curse of dimensionality, noise, and sparsity, which are common in high-dimensional data. The main techniques applied to the clustering ensemble process are explained below.

Singular Value Decomposition (SVD) factorizes a matrix into three components: left singular vectors, singular values, and right singular vectors. This decomposition enables low-rank approximations of the original matrix, which helps in enhancing pattern detection and highlighting dominant trends in the data [26]. SVD has been effectively utilized as a dimensionality reduction technique before clustering, particularly for handling sparse and high-dimensional datasets. However, SVD is computationally expensive, especially for large datasets, and its sensitivity to outliers and nonlinearities can sometimes degrade clustering performance [25].

Principal Component Analysis (PCA) is one of the most widely used linear dimensionality reduction techniques. PCA applies SVD in a specific way to maximize variance and identify principal components, transforming the original data into a set of orthogonal components called principal components, which capture the maximum variance in the dataset. PCA finds the optimal linear projections using eigenvector decomposition, ensuring minimal information loss [27]. Although PCA is utilized in clustering ensembles to reduce data dimensionality, its effectiveness depends on the dataset and clustering method. PCA combined with random subsampling (PCASS) has been studied for ensemble clustering, where it can improve clustering performance by generating diverse ensemble members, though its success may vary across datasets [19].

Random Projection (RP) is a dimensionality reduction technique that maps high-dimensional data onto a lower-dimensional subspace using a randomly generated projection matrix, while approximately preserving distances between data points. Theoretical guarantees, such as the Johnson-Lindenstrauss lemma [28], ensure that pairwise distances are maintained with minimal distortion. In ensemble clustering, random projections are valuable for generating diverse base clusterings by projecting the data onto different random subspaces, which enhances the robustness and quality of the final consensus clustering. These diverse views help capture various aspects of the data structure, mitigating issues like sparsity and high-dimensional noise [12].

D. Generation of Base Clusterings

The generation of base clusterings involves producing multiple clustering results that can later be combined into a consensus solution. The quality and diversity of the base clusterings are critical for the performance of the ensemble clustering method.

To generate diverse base clusterings, several strategies are commonly employed. One approach involves applying different clustering algorithms, such as k-means, hierarchical clustering, spectral clustering, or density-based clustering, to the same dataset to produce varied results. This method leverages the distinct strengths and criteria of each algorithm, enhancing ensemble diversity [29]. Another strategy focuses on parameter variations, where the same algorithm is run with different configurations, such as varying the number of clusters, initialization methods, or distance metrics. This ensures that even a single algorithm can yield diverse clusterings [13].

Subsampling and random projection offer another effective approach by utilizing random subsets of data points or projecting the data onto random subspaces to create diverse clustering outputs [6], [12], [30]. Additionally, “weak” clustering algorithms, which are simple methods, such as random 1-dimensional projections or random splitting by hyperplanes, that might not perform well individually, are often integrated into ensembles. Their contribution lies in providing diversity, enabling meaningful ensemble results [9], [10].

Ensemble clustering combines a variety of clustering algorithms to leverage their unique strengths and mitigate their individual limitations. Among these, partition-based methods like k-means and k-medoids play a prominent role due to their simplicity and efficiency. These algorithms produce hard partitions of the data, meaning each data point belongs to exactly one cluster. Their computational speed and ease of implementation make them widely used, especially in scenarios where the data conforms to spherical clusters or low-dimensional spaces. However, they may face challenges when applied to high-dimensional or complex data structures [31].

Hierarchical clustering, which constructs dendrograms via single-linkage, complete-linkage, or average-linkage algorithms, is commonly used as one of the base clustering methods in ensemble clustering frameworks. By generating base clusterings with hierarchical algorithms, potentially using different linkage criteria or varying subsets of features or data points, ensemble methods can capture a range of structural patterns and clustering tendencies present in the data. The diversity among base clusterings produced by hierarchical approaches contributes to the robustness and stability of the ensemble solution. However, hierarchical clustering can be computationally intensive for large-scale problems, especially when constructing dendrograms for large datasets [13].

Spectral clustering, such as the Ng–Jordan–Weiss method [32], offers a graph-based approach to partitioning data. By leveraging eigenvalue decomposition of similarity matrices, spectral clustering transforms the clustering problem into a graph partitioning challenge. This method is well-suited for

non-linearly separable data and excels in detecting clusters with intricate structures. Despite its powerful capabilities, spectral clustering requires careful handling of input parameters and computational resources, particularly for large-scale datasets [29].

Density-based methods such as Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise (DBSCAN) [33] and Ordering Points To Identify the Clustering Structure (OPTICS) [34] focus on identifying clusters based on varying densities. These algorithms are particularly adept at handling data with noise and detecting clusters of arbitrary shapes. Their reliance on density metrics allows them to effectively exclude outliers and uncover patterns in datasets where traditional methods struggle. However, selecting appropriate parameters for density thresholds can be challenging and requires domain expertise [13].

Fuzzy clustering methods, such as Fuzzy C-Means [35], provide a unique approach by allowing data points to belong to multiple clusters with varying degrees of membership. This is particularly useful for ambiguous or uncertain datasets where hard partitions may oversimplify the underlying structure. Fuzzy clustering is effective in capturing nuanced relationships between data points but can be computationally demanding, especially when dealing with high-dimensional data [13], [36].

Model-based clustering methods, such as Gaussian Mixture Models (GMMs), represent another important class of algorithms for generating base clusterings [6]. GMMs assume that data are generated from a mixture of several Gaussian distributions, each corresponding to a cluster, and use statistical methods (typically the Expectation-Maximization algorithm) to estimate the parameters of these distributions. This probabilistic approach allows for soft assignments of data points to clusters and can capture more complex cluster shapes than methods like k-means. GMMs are especially valuable in situations where clusters have different covariance structures or overlap in feature space, though they can be computationally intensive for high-dimensional data and require model selection strategies to determine the optimal number of components.

These diverse clustering algorithms, each with its strengths and weaknesses, can be integrated into the ensemble clustering process to enhance robustness, accuracy, and stability across varying data types and structures.

E. Ensemble Selection

Ensemble selection in clustering ensembles focuses on choosing the best subset of base clusterings to improve consensus results, since including all partitions, especially low-quality or redundant ones, can reduce performance. Selecting clusterings that are both diverse and of high quality can significantly improve the resulting consensus partitions [20].

High quality refers to base clusterings that are internally coherent and capture meaningful structure within the data; in practice, this is often assessed using internal validity measures or by evaluating how well a clustering agrees with the overall trend of the ensemble, such as through the Sum of Normalized

Mutual Information (SNMI). Diversity, on the other hand, measures how different the selected clusterings are from each other, ensuring that the ensemble combines complementary perspectives rather than redundant solutions. This is commonly quantified by calculating low pairwise similarity (e.g., low NMI) between clusterings [37].

Recent research has advanced clustering ensemble selection by introducing sophisticated strategies such as hierarchical and multiplex network-based approaches, which utilize ensemble diversity and quality through various validity indices and community detection techniques [38], [39]. Other works have developed selection strategies that explicitly combine internal and external validity measures to guide the process and improve consensus performance [40]. There is also a growing emphasis on adaptive, data-driven criteria for ensemble selection, where the contribution of each candidate partition is evaluated relative to the ensemble in a dynamic and context-aware manner [37]. Furthermore, recent studies highlight the benefits of ensemble selection for computational efficiency, particularly in large-scale and high-dimensional settings, where reducing ensemble size while maintaining accuracy is essential [13].

F. Consensus Functions

A core component of clustering ensemble methods is the consensus function—the strategy that combines multiple base clusterings into a single, robust consensus partition. The choice of consensus function significantly influences the quality, stability, and interpretability of the final clustering solution. Various consensus functions have been developed, which can be broadly categorized based on the type of information they utilize and their computational mechanism.

In the context of clustering ensembles, n denotes the number of data points or objects being clustered, r (also sometimes denoted as m or H) is the number of base clusterings (partitions) in the ensemble, and k is the number of clusters in the final consensus partition. Below, a taxonomy of consensus functions is presented [6], [10], [13], [17].

1) *Relabeling and Voting-Based Approaches*: These methods address the label correspondence problem by aligning cluster labels across partitions—typically using the Hungarian algorithm [41], which has a computational complexity of $O(k^3)$ —before assigning each object to the cluster label that receives the most votes (via plurality or majority voting). Such approaches are most effective for small datasets and for ensembles where the number of clusters is fixed across all partitions, allowing reliable and efficient label alignment. However, as the number or diversity of clusters increases, or when base clusterings contain different numbers of clusters, these methods become less practical due to the ill-posed nature and computational complexity of label alignment.

Majority voting is a simple and intuitive example: after label alignment, each data point is assigned to the cluster label it receives most frequently across all base clusterings. This method is most feasible when the number of clusters is

fixed and label alignment is reliable; otherwise, ambiguities and computational costs escalate.

The k-modes algorithm is another approach suited to categorical data and categorical cluster assignments. It generalizes k-means clustering to operate directly on categorical variables, grouping cluster assignments based on the mode (most frequent value) rather than the mean. This makes k-modes especially practical for consensus over cluster labels, which are inherently categorical.

2) *Co-association / Pairwise Similarity Matrix Methods*: These methods construct an $n \times n$ co-association matrix, where each entry (i, j) records the frequency (or proportion) with which objects i and j appear in the same cluster across ensemble members. This matrix serves as a new similarity measure, to which a standard clustering algorithm (such as hierarchical agglomerative clustering) is applied to extract the consensus clustering. Evidence Accumulation Clustering is a prominent example, using the co-association matrix as input for linkage-based clustering to produce the consensus partition. While intuitive and effective for small to medium datasets, this approach has $O(n^2)$ computational and memory complexity, making it unsuitable for very large data sets.

3) *Graph and Hypergraph-Based Methods*: These approaches represent relationships among data points and clusters as graphs or hypergraphs: nodes correspond to data points (or clusters), and edges or hyperedges encode co-membership or similarity. Consensus clustering is achieved by partitioning this (hyper)graph using graph partitioning algorithms.

The Cluster-based Similarity Partitioning Algorithm (CSPA) constructs a similarity (co-association) matrix where each entry reflects how often two data points are assigned to the same cluster across all partitions. This matrix is interpreted as a weighted adjacency graph, and a graph partitioning algorithm (such as spectral clustering) is applied to derive the consensus partition. CSPA is effective but has a quadratic time and space complexity in the number of data points.

The Linkage Clustering Ensemble (LCE) method constructs a link-based similarity between clusters across partitions, capturing not just pairwise object similarity but also higher-order relationships. It then applies clustering to these links to produce the consensus. LCE is particularly useful when cluster relationships are more informative than direct object co-occurrence, and has been shown to perform well in various applications.

4) *Model-Based and Optimization Methods*: These methods formulate consensus clustering as an optimization or statistical inference problem, aiming to find the consensus partition that best explains or summarizes the ensemble.

Latent Class Analysis (LCA) is a probabilistic approach that models the observed cluster assignments as being generated by latent (hidden) class variables. LCA infers the most likely consensus cluster labels by estimating the underlying probability distributions, making it well-suited for situations where partitions are noisy or partially observed.

Other optimization-based approaches, such as those using Non-Negative Matrix Factorization (NMF), encode the ensem-

ble into a matrix and factorize it to yield soft assignments, directly targeting the consensus objective. These are theoretically well-founded and efficient for large datasets but can be computationally demanding for very large n or complex similarity metrics.

G. Validation Metrics and Evaluation

Evaluating the quality and effectiveness of clustering ensembles is a fundamental task, especially in the absence of ground truth labels, a common scenario in unsupervised learning. The assessment of clustering solutions typically relies on cluster validity indices, which can be broadly categorized into internal, external, and relative indices. Each of these metrics evaluates different aspects of clustering performance and offers complementary insights into the quality of the partitions produced by ensemble methods [13].

Internal validity indices are essential tools for evaluating the quality of clustering solutions without relying on external information such as class labels. These indices assess clustering results by examining the inherent structure of the data, focusing primarily on two key aspects: compactness (how closely related the data points within the same cluster are) and separation (how distinct different clusters are from each other).

Some of the most widely used internal validity indices are the Davies-Bouldin Index, which evaluates the average similarity between each cluster and its most similar one, rewarding compact and well-separated clusters with lower values; the Silhouette Coefficient, which combines measures of both intra-cluster cohesion and inter-cluster separation, with higher average values indicating better clustering; and the Calinski-Harabasz Index and Dunn Index, which rely on ratios of between-cluster and within-cluster distances to quantify cluster definition, each with its own sensitivity to cluster shapes and densities. Other metrics, like the C-Index, Gamma Index, PBM Index, Ray-Turi Index, and McClain-Rao Index, introduce alternative formulations for assessing the internal structure of clusters, often considering variances, pairwise distances, or cluster compactness [37].

Since ground truth labels are typically unavailable, internal clustering validity metrics are commonly used for unsupervised model selection. However, different metrics may favor different clustering characteristics and sometimes produce conflicting recommendations. To overcome this, rank aggregation methods can be employed to combine multiple internal indices into a single consensus ranking. The weighted rank aggregation approach [42], based on the Monte Carlo Cross-Entropy (CE) algorithm, optimizes a weighted Spearman's footrule distance to objectively synthesize rankings from various validation measures. This automated method is especially useful when comparing many algorithms or cluster configurations, providing a robust and holistic assessment of clustering quality.

External validity indices compare clustering results to an external reference or ground truth (when available), providing a direct measure of agreement between the obtained clusters

and known class labels. A wide range of external indices have been developed, each capturing different aspects of similarity or agreement. Commonly used indices include the Adjusted Rand Index (ARI) and the Rand Index (RI), which are pair-counting measures that quantify the proportion of sample pairs on which two partitions agree; the Normalized Mutual Information (NMI), Mutual Information (MI), and Adjusted Mutual Information (AMI), which are information-theoretic measures evaluating the shared information between cluster assignments and ground truth; and the Variation of Information (VI) and Entropy, which quantify the information lost or gained between the two partitions.

Set-matching measures such as Purity and F-Measure (F1-score) assess the extent to which clusters contain a single class or the harmonic mean of precision and recall, respectively. Additional pairwise indices like the Jaccard Index and the Fowlkes-Mallows Index also provide valuable perspectives on clustering agreement. More recently, the Chi Index, based on the chi-squared statistical test, has been proposed to directly assess the dependence between clustering and ground truth assignments using contingency tables. [37], [43]

Relative validity indices are designed to compare the quality of different clustering results or parameter settings on the same dataset by quantifying how changes in the clustering structure affect certain quality measures. Unlike absolute indices, which provide an overall assessment of a single clustering, relative indices are particularly useful for evaluating the impact of algorithmic choices, such as the number of clusters, distance metrics, or initialization methods. [37]

Empirical validation in clustering ensemble research frequently relies on combining multiple validity indices to achieve a comprehensive evaluation, particularly for high-dimensional datasets where individual metrics may be inadequate. Benchmarking on both synthetic and real-world datasets, in conjunction with the use of diverse validity indices, is recommended to assess the reliability and generalization of ensemble clustering methods.

III. SOLUTION PROPOSAL

This solution employs ensemble clustering tailored for high-dimensional datasets to enhance robustness, stability, and accuracy. By integrating multiple base clustering results into a unified consensus partition, this approach leverages the strengths of diverse clustering methods, effectively addressing challenges such as sparsity, noise, and the lack of a universally optimal individual clustering method. This methodology was specifically chosen for its suitability to high-dimensional settings and its ability to deliver strong results even under limited computational resources, without relying on complex or highly resource-intensive algorithms.

Fundamentally, the method relies on two key principles: diversity in base clusterings, achieved by varying algorithms, parameters, and feature subsets, and a consensus function, responsible for synthesizing these heterogeneous partitions into a single, cohesive clustering solution.

The six key steps of the proposed solution, as illustrated in Figure 2, are as follows:

1) **Preprocessing and Unsupervised Feature Selection:**

First, standard data preprocessing such as normalization and handling missing values is performed. Then, Infinite Feature Selection (Inf-FS) algorithm is employed as an unsupervised feature selection method. This technique is selected because it automatically determines the optimal number of features to retain, thereby facilitating reproducibility and reducing manual intervention. Feature selection is applied as a first step to mitigate the curse of dimensionality and improve cluster structure. However, it is important to note that in datasets with low redundancy—where most features carry unique, non-overlapping information—feature selection may not be appropriate, as removing features could result in loss of critical information relevant for clustering.

2) **Random Projections or Subsampling**

A set of B Gaussian random projections is generated from the feature subset selected by Inf-FS. Among these, the B^* most informative projections are selected based on their Bayesian Information Criterion (BIC) scores, ensuring that only the most statistically coherent projections are retained. This strategy enhances the scalability of the ensemble clustering framework, increases structural diversity among base clusterings, and improves robustness to noise and redundancy.

The projected dimension d of each random projection is determined based on theoretical and empirical findings regarding the preservation of cluster structure. In particular, projecting high-dimensional features onto a subspace of dimension $d = O(\log G)$, where G is the number of clusters, is sufficient to almost perfectly preserve inter-cluster separation for GMMs. The empirical analysis further recommends $d = 10\log(G) + 1$ as a practical guideline, noting that increasing d beyond this value does not significantly improve clustering performance [12].

In datasets where redundancy is low and most features are representative, an alternative to random projections for creating diversity is the subsampling-based approach inspired by the consensus clustering methodology introduced by [8]. This method generates multiple perturbed datasets from the original data preserving the intrinsic structure while introducing controlled variability. Each perturbed dataset is clustered independently and the results are aggregated into a consensus matrix that quantifies the stability and reproducibility of clusters across multiple runs.

3) **Diverse Clustering Algorithms:**

To generate the base clusterings, a selected set of 5 algorithms is used, representing key methodological families. These included k-means (KM), a classic centroid-based method effective for well-separated, spherical clusters; Gaussian Mixture Models (GMM), a probabilistic ap-

proach suitable for overlapping distributions; Spectral Clustering (SC), which leverages graph theory to detect complex structures; Fuzzy c-means (CMEANS), enabling soft assignments where data points can belong to multiple clusters; and Partitioning Around Medoids (PAM), a medoid-based technique known for its robustness to outliers. This selection ensures diverse clustering perspectives by combining geometric, probabilistic, graph-based, and soft-partitioning strategies.

4) **Validation Metrics and Evaluation:**

For each dataset, the best performing combination of feature selection, projection, clustering algorithm, and consensus function is identified. The evaluation is based on external metrics most widely accepted in the literature for that particular dataset (e.g., Adjusted Rand Index, Normalized Mutual Information, accuracy, etc.), ensuring fair and meaningful comparisons.

5) **Statistical Significance Analysis:**

The statistical significance of the performance improvement is assessed. Specifically, the best ensemble configuration is compared against the other configurations. Since the executions are not strictly paired—due to the changes introduced by the feature selection and/or random projections in the pipeline—the appropriate statistical test is the Mann-Whitney U test (also known as the Wilcoxon rank-sum test) for independent samples. This is a non-parametric test, meaning it does not assume that the data follow a normal distribution.

Since clustering is typically performed in settings where ground truth labels are unavailable, all configuration choices within the ensemble framework are made in a fully unsupervised manner. At each stage—such as feature selection, dimensionality reduction, clustering algorithm selection, and consensus function choice—only internal clustering validity metrics guide decision-making. Multiple metrics (e.g., Silhouette, Davies-Bouldin, Calinski-Harabasz) are computed for every configuration, and an aggregate ranking procedure is applied: configurations are individually ranked by each metric, and their combined score determines the overall ranking. This work, for example, applies a weighted rank aggregation strategy based on Monte Carlo Cross-Entropy [Pihur 2007]. The configuration with the top aggregate rank is selected as optimal, independently of any external labels. External metrics are used only for post-hoc evaluation and reporting.

A. Experimental Framework

The experimental framework was designed to ensure reproducibility, scalability, and extensibility across different computational environments and programming languages.

1) *Computing Setup:* The experiments were conducted on a computer running Microsoft Windows 11 Home, equipped with an Intel Core i9-14900HX processor (32 cores) and 32 GB of RAM.

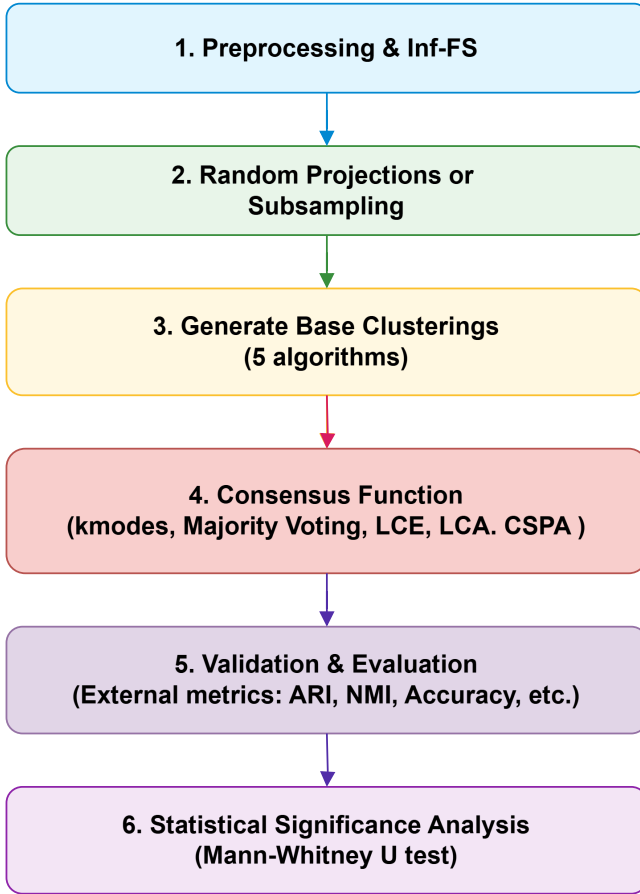


Fig. 2. Pipeline for the proposed clustering ensemble method

2) *Programming Tools and Libraries*: The diceR package [44], [45] in R served as the primary tool for implementing the ensemble clustering framework. This package provides a comprehensive suite of functions for performing cluster analysis using an ensemble clustering approach. diceR applies the consensus clustering methodology through an ensemble of clustering algorithms combined with subsampling, aligning with the principles described in [8]. It extends the original consensus clustering approach by integrating multiple algorithms and consensus functions to improve clustering robustness. It supports the generation of diverse base clusterings through multiple algorithms, the application of various consensus functions, and the evaluation of clustering performance via both internal and external validity indices. To further expand its capabilities, an enhanced version called diceRplus [46] was developed, incorporating unsupervised feature selection, dimensionality reduction via random projections, ensemble algorithm RPEClu_parallel, and additional functionalities.

Auto-UFSTool [47] was utilized for unsupervised feature selection. This MATLAB toolbox provides a collection of 25 robust unsupervised feature selection approaches, most of which were developed within the last years. It enables the evaluation of feature selection results and generates comparative graphs for different feature subsets. By incorporating Auto-

UFSTool into the workflow, the ensemble clustering framework can benefit from state-of-the-art unsupervised feature selection techniques, potentially improving the quality and relevance of the input features for subsequent clustering steps.

R.matlab [48] was used to connect R and MATLAB. It is an R package that allows communication between R and MATLAB via a TCP/IP connection. The package provides methods for reading and writing MAT files, as well as sending commands and data between R and MATLAB.

3) *Experimental Design and Reproducibility*: To guarantee reproducibility and systematic analysis, an experiment management framework was established. This included:

- **Configuration Management**: All experiment configurations (algorithm choices, parameter settings, random seeds) were stored in version-controlled files.
- **Result Storage**: Outputs—including cluster assignments and validity scores—were archived in structured format (dataframe) for traceability and post-hoc analysis.
- **Automation & Analytics**: Scripts were developed to automate the running of experiments, aggregation of results, and generation of analytical visualizations.
- **Documentation and Code Sharing**: All scripts and documentation were maintained in a Git repository [46] to promote transparency and reusability.

IV. EXPERIMENTAL RESULTS AND EVALUATION

This section presents the evaluation of the proposed ensemble clustering method using several benchmark high-dimensional datasets. These datasets come from various domains, such as spectroscopy, genomics, and image analysis, and are used to assess the generalizability and robustness of the ensemble clustering framework. Table I summarizes the main characteristics of the datasets, including their domain, a brief description, and their dimensionality.

In the experiments, the number of clusters k was fixed for each dataset based on prior knowledge or reference values. This decision was made to reduce experimental complexity and computational load, keeping the number of configurations manageable and ensuring reproducibility. Although not all clustering algorithms require a predefined number of clusters, fixing k allows a fair and consistent comparison.

A. Initial Experiments: Validating the Integration of Unsupervised Feature Selection with Ensemble Clustering

To validate the proposed methodology, a set of initial experiments was designed to assess the impact of unsupervised feature selection when integrated with ensemble clustering. The focus was on evaluating how feature selection influences clustering performance, particularly in combination with the random projection-based RPEClu algorithm [12]. The goal was to examine whether removing irrelevant or redundant features before clustering could help reduce sensitivity to noise and lower computational costs, two common challenges in high-dimensional data analysis.

TABLE I
SUMMARY OF BENCHMARK DATASETS USED IN EXPERIMENTS

Dataset	Domain	Description	Samples	Features
ALLAML [50]	Genomics	Leukemia mass spectrometry data.	72	7,129
Glioma [51]	Genomics	Glioma tumors (brain or spinal cord cancers).	50	4,434
Lung Cancer [52]	Genomics	Expression profiles of lung cancer types and normal tissue.	203	3,313
Lymphoma [53]	Genomics	Lymphoma gene expression.	62	4,026
Meat [54]	Food	Spectroscopy of raw meat from 5 animal species.	231	1,050
Prostate GE [55]	Genomics	Microarray data from prostate tumor and normal samples.	102	5,966
TOX-171 [51]	Genomics	Microarray data from prostate tumor and normal samples.	171	5,748
warpPIE10P [51]	Image	PIE face subset.	210	2,420

In this setup, the RPEClu approach applies random projections to map the original high-dimensional data into lower-dimensional subspaces. Within each projected subspace, GMM clustering is performed, and the resulting clusterings are aggregated through consensus functions. Infinite Feature Selection (Inf-FS) was used to rank and select the most informative features before the ensemble step. To improve efficiency, a parallel version of RPEClu was implemented, allowing random projections to run concurrently and significantly reducing execution time while preserving clustering quality.

1) *RPEClu: Original vs. Parallelized Implementation:* The single-threaded implementation of RPEClu ($G = 5, d = 17, B = 1000, B^* = 100$)—as reported in [12]—was applied to the Meat dataset for replication. A parallelized version was developed to optimize performance. As shown in Table II, execution time was greatly reduced with minimal change in ARI, confirming the efficiency and reliability of the updated method.

TABLE II
RPECLU IMPLEMENTATION PERFORMANCE

RPEClu Implementation	Exec. Time (s)	ARI
Single-threaded as reported in [12]	6,460	0.32
Single-threaded version	2,475	0.29
Parallelized version	356	0.31

2) *Impact of Unsupervised Feature Selection:* Experiments were conducted using Inf-FS between 25 and 1,050 variables, increasing in steps of 25. RPEClu algorithm ran with ($G = 5, d = 17$) and 2 configurations ($B = 1000, B^* = 100$ and $B = 500, B^* = 50$). Internal clustering metrics, as a function of the number of features selected by Inf-FS, are summarized in Figure 3.

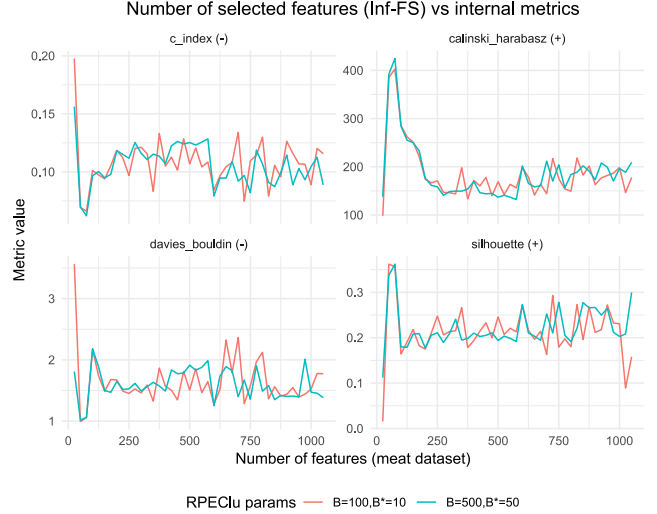


Fig. 3. Internal clustering metrics (y-axis) as a function of the number of features selected by Infinite Feature Selection (x-axis) on the Meat dataset, evaluated using RPEClu with two parameter configurations. Higher (+) or lower (-) values indicate better clustering.

These plots illustrate how the performance of RPEClu varies across a range of feature subset sizes, from 25 up to 1,050 variables. Notably, several internal metrics—including c-index, Calinski-Harabasz, Davies-Bouldin and silhouette—reach optimal or near-optimal values at intermediate feature counts (typically between 50 and 100 features), suggesting that excessive dimensionality reduction may eliminate informative variables, while retaining too many features reintroduces noise and redundancy. Overall, these results indicate that careful tuning of the feature selection step is crucial for maximizing clustering performance: an appropriate balance between dimensionality reduction and information preservation leads to more compact, well-separated, and robust clusters. This trend is consistent across both parameter configurations tested ($B = 1000, B^* = 100$ and $B = 500, B^* = 50$), underscoring the stability of the ensemble clustering approach when combined with unsupervised feature selection.

Table III summarizes the best values achieved for each internal clustering metric across the different experimental configurations. Each entry indicates the experimental parameters, the corresponding number of selected features, and the best metric value obtained. Table IV presents a ranking of the experimental configurations that achieved the best values across the various internal validation indices, indicating for each configuration the number of metrics in which it attained the top result as well as the corresponding ARI value. This

table allows for the identification of the most robust configurations in terms of consistent performance according to multiple clustering quality metrics. Experiment with parameters ($B = 500, B^* = 50, \text{num. features} = 75$) stands out as the preferred configuration due to its superior consistency and robustness, achieving top results across four internal metrics. This demonstrates its ability to maintain high clustering quality across diverse evaluation criteria, ensuring stable and reliable outcomes in different scenarios. Additionally, its external metric remains consistently strong ($ARI = 0.68$)

TABLE III
BEST INTERNAL METRICS (INF-FS FEATURE SELECTION - MEAT DATASET)

Metric	Params (B, B^*)	Num Features	Best Value
C-index	500, 50	75	0.06
Calinski-Harabasz	500, 50	75	424.63
Compactness	1000, 100	25	0.13
Connectivity	500, 50	975	64.48
Davies-Bouldin	1000, 100	50	1.01
Dunn	500, 50	600	0.06
Gamma	500, 50	400	10.33
McClain-Rao	500, 50	75	0.27
pbm	500, 50	1050	13.53
Ray_Turi	1000, 100	50	0.52
Sd distr.	1000, 100	25	38.47
Silhouette	500, 50	75	0.36
Tau	500, 50	575	0.00

TABLE IV
RANKING OF BEST INTERNAL METRICS (INF-FS FEATURE SELECTION - MEAT DATASET)

Rank	Params (B, B^*)	Feat.	Count	ARI
1	500, 50	75	4	0.68
2	1000, 100	25	2	0.53
3	1000, 100	50	2	0.69
4	1000, 100	75	1	0.69
5	500, 50	400	1	0.44
6	500, 50	575	1	0.61
7	500, 50	600	1	0.37
8	500, 50	675	1	0.54
9	500, 50	975	1	0.30
10	500, 50	1050	1	0.31

Figure 4 illustrates the relationship between the number of selected features by Inf-FS and the ARI in the meat dataset. The baseline $ARI = 0.32$ reference line marks the ARI achieved without feature selection in [12], providing a benchmark to assess improvements across varying feature subsets. The plot shows substantial gains in clustering performance when feature selection is applied. Both configurations achieve comparable clustering results, but the $B = 500, B^* = 50$ setting offers a reduction in execution time, making it the more efficient choice.

B. Experimental results of the ensemble method in benchmark datasets

After establishing the effectiveness of integrating unsupervised feature selection with ensemble clustering in the initial validation experiments, this section systematically evaluates the ensemble clustering solution, as described in section III.

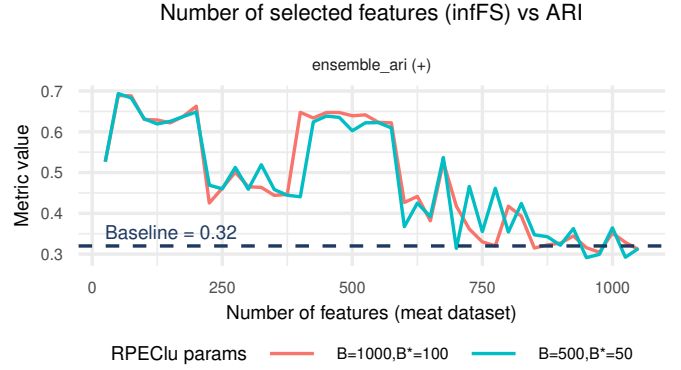


Fig. 4. Number of selected features (Inf-FS) vs ARI. Higher ARI values suggest better agreement between the clustering and ground-truth labels.

This phase involves benchmarking multiple combinations of feature selection, RPEClu and random projections, subsampling, clustering algorithms, and consensus functions across a range of high-dimensional datasets. Due to the exploratory nature of the experiments—focused on trend identification and comparative insight rather than formal hypothesis testing—statistical significance analysis is not conducted here.

RPEClu algorithm selects the top $B^* = 50$ projections from an initial set of $B = 500$, generated using features retained by the feature selection process. When RPEClu is not employed, diversity among base clusterings is introduced via subsampling: 80% of the data is randomly selected over 10 iterations, following the strategy described in [8].

Table V summarizes the performance of the proposed ensemble clustering configurations across all datasets. Each configuration combines different components, including the application or omission of feature selection via Inf-FS, dimensionality reduction through RPEClu, the choice of clustering algorithm, and the consensus function. Clustering and consensus functions were selected using a weighted aggregate ranking function [42] based solely on internal clustering validity metrics, ensuring that all decisions remain fully unsupervised. External evaluation metric ARI is computed only post hoc, for reporting and comparison purposes.

TABLE V
ARI FOR DIFFERENT CONFIGURATIONS. FEATURE COUNT SHOWN AS ORIGINAL/SELECTED (INF-FS).

Dataset	$\bar{r}$	k	Feats.	NoFS	FS	RPEClu
ALLAML	0.16	2	7,129/858	0.24	0.02	-0.01
Glioma	0.34	4	4,434/711	0.37	0.13	0.05
Lung Cancer	0.15	5	3,313/282	0.48	0.21	0.40
Lymphoma	0.21	3	4,026/469	0.85	0.79	1.00
Meat	0.84	5	1,050/91	0.24	0.33	0.69
Prostate GE	0.41	2	5,966/949	0.03	0.00	0.02
TOX-171	0.15	4	5,748/699	0.24	0.16	0.08
warpPIE10P	0.41	10	2,420/382	0.11	0.16	0.54

Feature count shown as original/selected (Inf-FS). ARI scores are reported for configurations of clustering and ensemble methods without feature selection (No FS), with feature selection (FS), and using RPClu. Mean correlation $\bar{r}$ was computed as the average absolute pairwise Pearson correlation among all features in each dataset

The table presents a comparative analysis of clustering performance across multiple ensemble configurations and datasets, with a focus on feature dimensionality before and after applying Inf-FS. Lymphoma continues to demonstrate robust separability, achieving near-perfect ARI in all conditions—most notably with RPClu. Similarly, warpPIE10P and Meat benefit substantially from RPClu’s dimensionality reduction, outperforming both NoFS and FS variants. In contrast, datasets like ALLAML and Prostate GE remain challenging regardless of configuration, reflecting either high intrinsic noise or subtle cluster structures that unsupervised methods struggle to capture. Interestingly, Feature Selection does not consistently improve results; in fact, it leads to substantial degradation on several datasets, underscoring the delicate balance between reducing dimensionality and preserving discriminative information.

While Inf-FS achieves a significant drop in dimensionality (e.g., from 7,129 to 858 in ALLAML), this reduction does not always translate to improved clustering—suggesting that aggressive filtering may omit relevant structure. Consensus clustering methods (such as RPClu) offer strong performance when the dataset has moderate internal consistency, as seen in Lung Cancer and Glioma. However, across all methods, variability in ARI scores highlights that no single approach is universally superior. These results collectively emphasize the importance of tailoring unsupervised pipelines to the statistical properties of each dataset.

V. CONCLUSIONS

This study presents an ensemble clustering framework tailored to address the challenges of high-dimensional data, including sparsity, noise, and feature redundancy. The framework integrates unsupervised feature selection (via Infinite Feature Selection, Inf-FS), dimensionality reduction through random projections (RPClu), subsampling to introduce diversity, and a varied set of clustering algorithms and consensus functions. Systematic benchmarking across multiple high-dimensional datasets shows that ensemble methods significantly enhance clustering robustness, stability, and accuracy, as reflected in improved external validity metrics.

Experimental results showed that the effectiveness of feature selection and dimensionality reduction is highly data-dependent. In datasets with high feature redundancy, combining feature selection and random projection with ensemble clustering led to significant performance gains. In contrast, in datasets with low redundancy, these steps offered limited benefits; in some cases, subsampling or leveraging algorithmic diversity alone was sufficient to achieve strong results. These findings underscore the importance of adaptive ensemble design tailored to the intrinsic characteristics of each dataset.

The experiments also highlighted the value of combining multiple clustering algorithms and consensus functions to leverage their complementary strengths. The proposed reproducible experimental framework enables systematic evaluation and comparison of ensemble configurations, enabling informed selection of optimal strategies for different scenarios.

Future work should explore automated selection and dynamic tuning of ensemble components, leveraging advanced approaches such as meta-learning, optimization heuristics, and reinforcement learning to adaptively configure feature selection, dimensionality reduction, and clustering strategies to dataset-specific characteristics. Incorporating additional feature selection methods tailored to the statistical properties of the data, as well as developing advanced consensus functions—potentially leveraging deep learning for complex data structures—remains a promising direction.

Another critical aspect will be the integration of explainable AI techniques to enhance the interpretability and transparency of clustering results. This will enable users to better understand and trust the discovered structures, supporting both scientific and industrial applications.

Further improvements in scalability will also be critical to support large-scale and streaming data scenarios. Expanding experimental evaluations to include a wider range of datasets—with more instances, increased domain diversity, and varying data profiles—will be important for assessing the generalizability and robustness of the proposed framework, especially under more diverse and challenging real-world conditions.

Lastly, future research should consider automated, data-driven methods for ensemble selection and parameter tuning, and continue to develop more effective visualizations and interpretation tools to maximize the practical impact of ensemble clustering solutions in high-dimensional settings

In summary, while no single ensemble configuration universally outperforms others, the flexible and adaptive use of feature selection, dimensionality reduction, and diverse clustering models within an ensemble framework offers a robust approach to high-dimensional data clustering. The results and codebase provided in this study lay a foundation for further advancements in this domain.

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